higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

WELDING
NQF LEVEL 3

(6031033)

11 March 2019 (X-Paper)
09:00–12:00

Calculators may be used.

This question paper consists of 5 pages.

NCV

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

NCV

QUESTION 1

Choose an item from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.1–1.10) in the ANSWER BOOK.

COLUMN A		COLUMN B	
1.1	Process to make slabs of steel	A	extrusion
1.2	Contains iron and carbon	B	hot working
1.3	Smallest distance across a weld	C	density × volume
1.4	Low-carbon steel	D	bevelling
1.5	Mass of material	E	throat
1.6	Fillet weld	F	drop forging
1.7	Shaping of metals at high temperatures	G	0,05 to 0,3%
1.8	Pushing metal through a shaped die	H	T-piece
1.9	Dropping a large hammer on a workpiece	I	continuous casting
1.10	Increasing the strength of a weld	J	non-alloy steels

(10 × 1) **[10]**

QUESTION 2

2.1 List SIX important effects of flux on the welding process. (6)

2.2 Give FIVE ways to eliminate welding hazards in accordance with standard working procedures. (5 × 2) (10)

2.3 Explain why it is important to report faults on tools and equipment. (4)

2.4

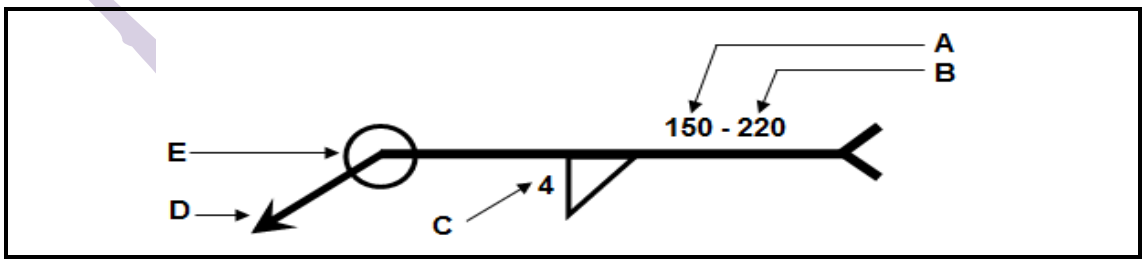


FIGURE 1

Label the welding symbols shown in FIGURE 1 by writing only the answer next to the letter (A–E) in the ANSWER BOOK. (5 × 1) (5)

[25]

QUESTION 3

3.1 List the components that make up an oxy-acetylene welding apparatus. (8)

3.2 List THREE positions that can be done with oxy-acetylene welding. (3)

3.3 List NINE welding defects that can be found in oxy-acetylene welding. (9)

[20]

QUESTION 4

4.1

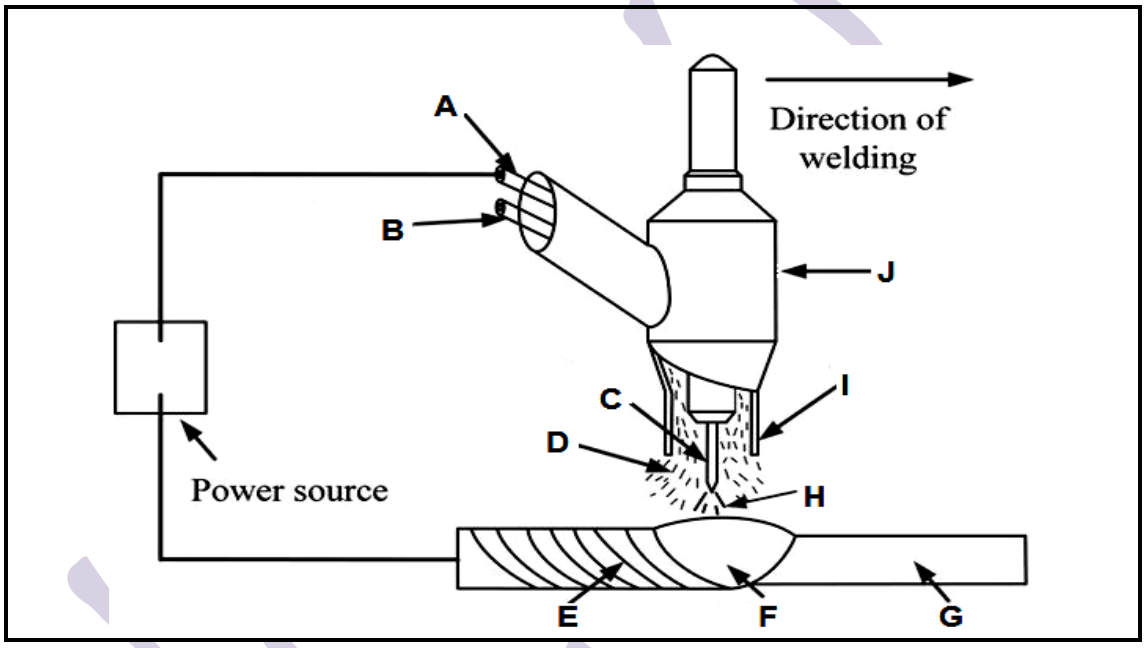


FIGURE 2

Label the diagram of a gas tungsten arc-welding process shown in FIGURE 2 by writing only the answer next to the letter (A–J) in the ANSWER BOOK.

(10 × 1) (10)

4.2 Write each of the following abbreviations in full:

4.2.1 DCEN

4.2.2 DCEP

4.2.3 AC

(3 × 1) (3)

4.3 List SEVEN safety measures to prevent electric shock. (7)

4.4 List THREE ways to prevent getting burnt when working with hot workpieces. (3)

4.5 State TWO functions of a flowmeter regulator. (2)

[25]

QUESTION 5

5.1 Give FIVE primary precautions to prevent the inhalation of harmful fumes and gases. (5)

5.2 State TWO requirements when selecting welding cables. (2)

5.3 State the functions of an electrode wire. (2)

5.4 Calculate the amperage required to weld a 10 mm plate.

Hint: 0,0254 mm = 1 A (5)

5.5 List SIX checks to comply with a work-area inspection. (6)
[20]

TOTAL: 100

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